# Kernix 0.9.0 / Performance brief

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

**25** workloads

**100** observations

Source precision retained

## Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product




## How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

# Kernix 0.9.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

## Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product




## Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



# Kernix 0.9.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

## Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product




## Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



# Kernix 0.9.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

## Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product




## Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product

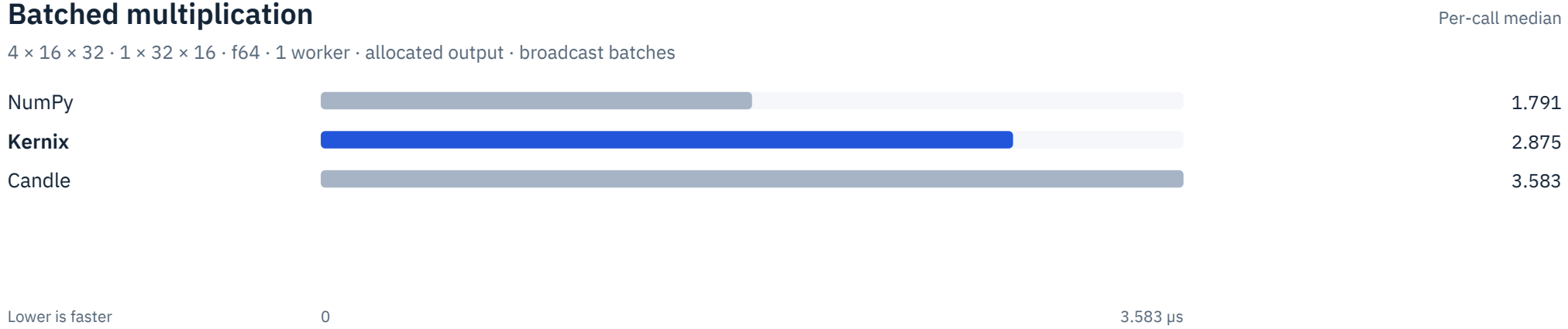


# Kernix 0.9.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

## Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



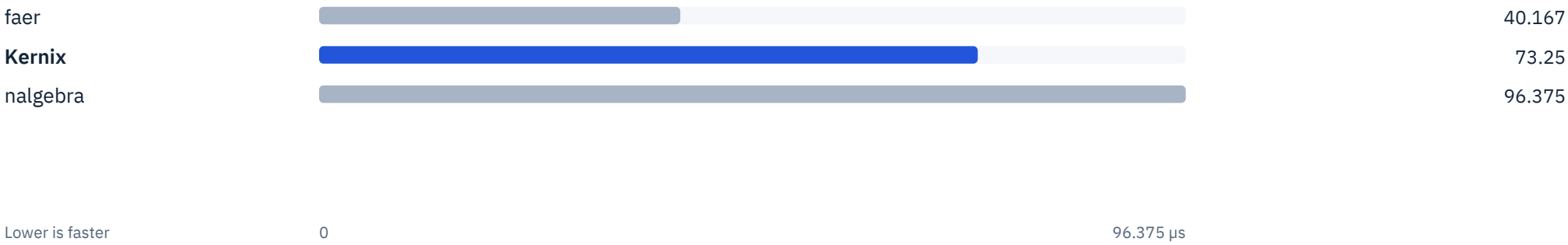
# Kernix 0.9.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

## Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

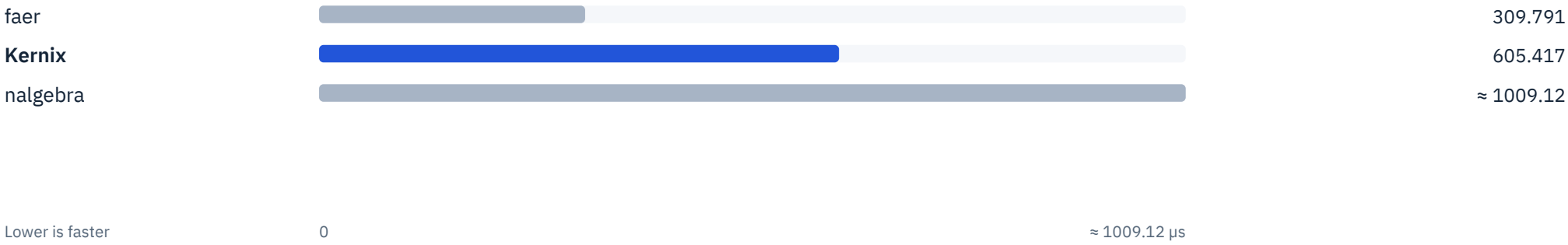
Per-call median




## Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



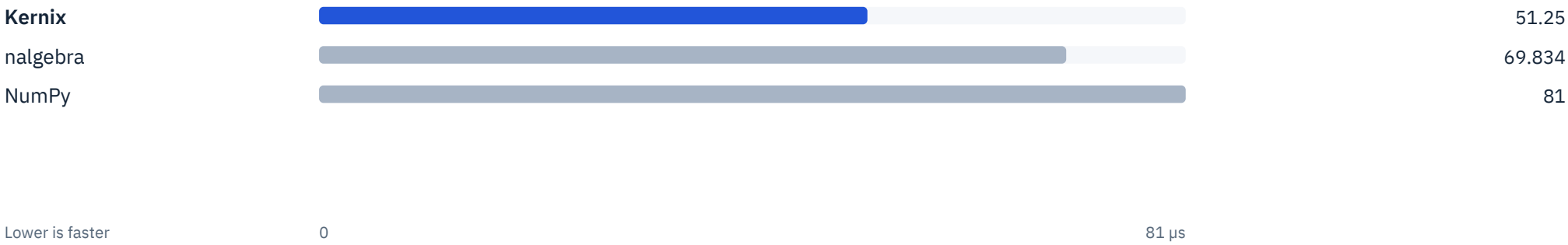
# Kernix 0.9.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

## Minimum-norm least squares

Per-call median

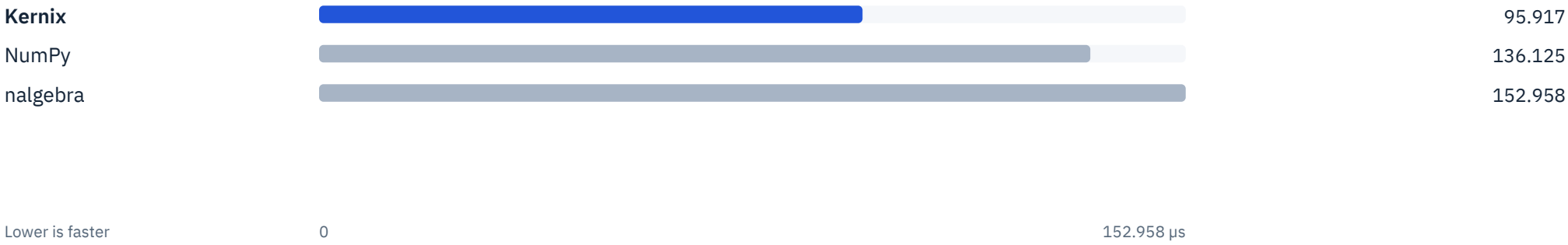
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ




## Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



# Kernix 0.9.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

## Thin QR

96 × 32 · f64 · 1 worker · allocated output




## Thin QR

256 × 64 · f64 · 1 worker · allocated output



# Kernix 0.9.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

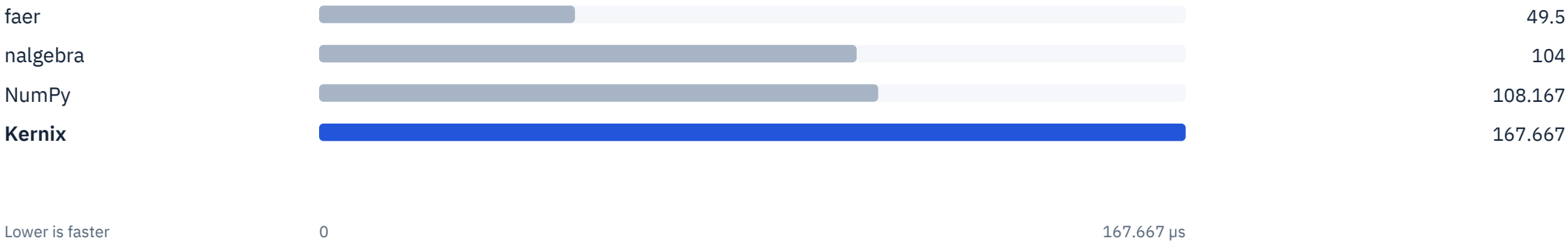
## Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output




## Cholesky factorization

128 × 128 · f64 · 1 worker · allocated output

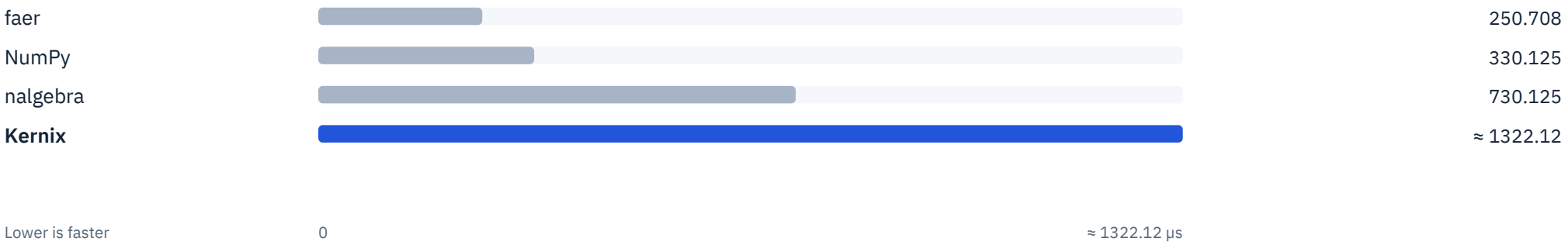


# Kernix 0.9.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

## Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



# Kernix 0.9.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

## Record + backward

Per-call median

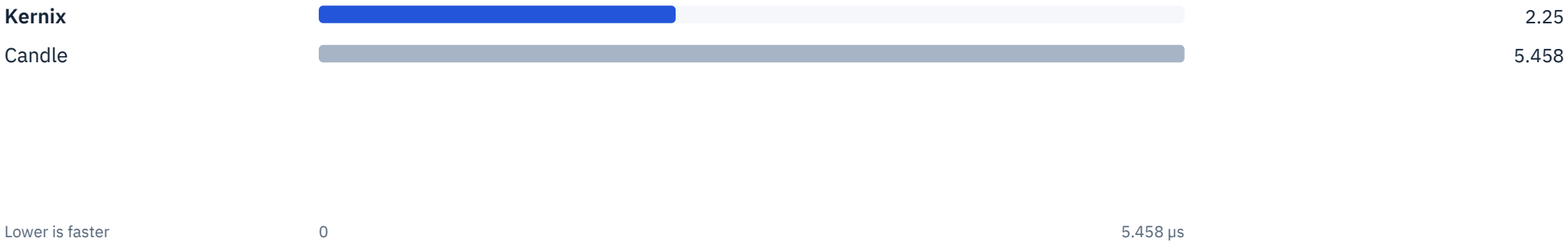
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product




## Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



# Kernix 0.9.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

## Model recording + forward

Per-call median

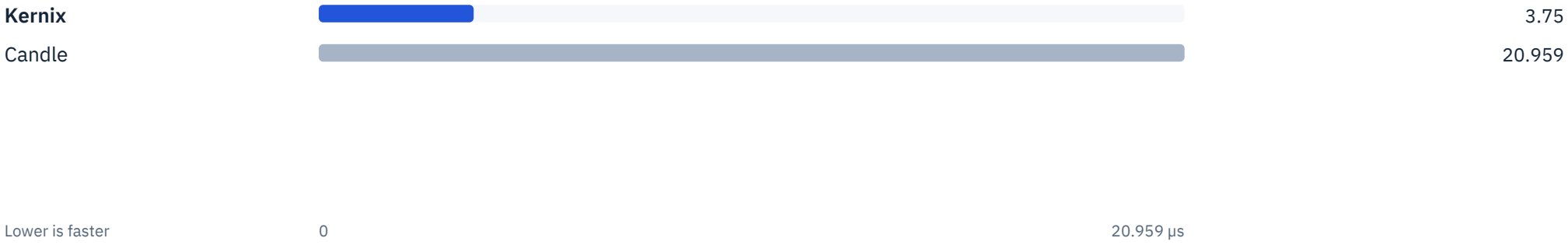
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented




## Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



# Kernix 0.9.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

## Parameter update

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

Per-call median



# Kernix 0.9.0 / Statistics & activations

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

## Row softmax

256 × 64 · f64 · 1 worker · allocated output



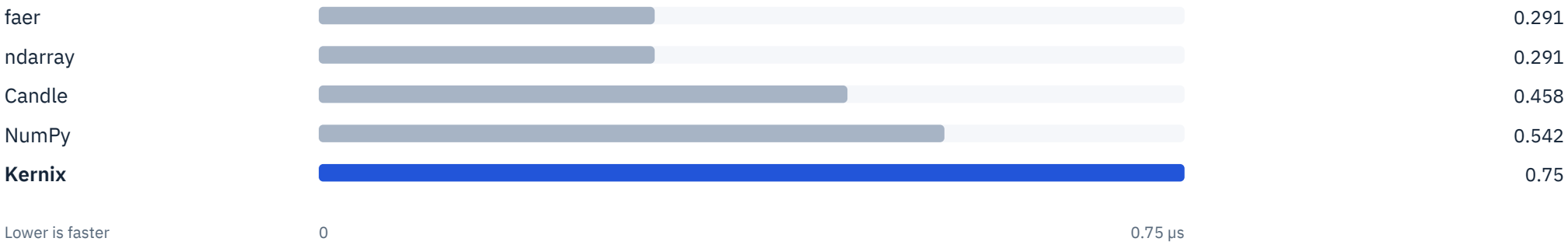
# Kernix 0.9.0 / Tensor construction

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

## Create zeros

64 × 64 · f32 · 1 worker · allocated output

Per-call median




## Create ones

64 × 64 · f32 · 1 worker · allocated output

Per-call median



# Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / Kernix**<br>32 × 32 · 32 × 32 · f64 | 2250 | 2.25 |
| **Matrix multiplication / faer**<br>32 × 32 · 32 × 32 · f64 | 2417 | 2.417 |
| **Matrix multiplication / ndarray**<br>32 × 32 · 32 × 32 · f64 | 2083 | 2.083 |
| **Matrix multiplication / Candle**<br>32 × 32 · 32 × 32 · f64 | 2583 | 2.583 |
| **Matrix multiplication / NumPy**<br>32 × 32 · 32 × 32 · f64 | 1541 | 1.541 |
| **Matrix multiplication / nalgebra**<br>32 × 32 · 32 × 32 · f64 | 2083 | 2.083 |
| **Matrix multiplication / NumPy**<br>64 × 64 · 64 × 64 · f64 | 3500 | 3.5 |
| **Matrix multiplication / nalgebra**<br>64 × 64 · 64 × 64 · f64 | 15583.999999999998 | 15.583999999999998 |
| **Matrix multiplication / Kernix**<br>64 × 64 · 64 × 64 · f64 | 12959 | 12.959 |
| **Matrix multiplication / ndarray**<br>64 × 64 · 64 × 64 · f64 | 13375 | 13.375 |

# Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / faer**<br>64 × 64 · 64 × 64 · f64 | 14709 | 14.709 |
| **Matrix multiplication / Candle**<br>64 × 64 · 64 × 64 · f64 | 14500 | 14.5 |
| **Matrix multiplication / faer**<br>128 × 128 · 128 × 128 · f64 | 108500 | 108.5 |
| **Matrix multiplication / NumPy**<br>128 × 128 · 128 × 128 · f64 | 17667 | 17.667 |
| **Matrix multiplication / nalgebra**<br>128 × 128 · 128 × 128 · f64 | 103667 | 103.667 |
| **Matrix multiplication / ndarray**<br>128 × 128 · 128 × 128 · f64 | 109792 | 109.792 |
| **Matrix multiplication / Kernix**<br>128 × 128 · 128 × 128 · f64 | 102541 | 102.541 |
| **Matrix multiplication / Candle**<br>128 × 128 · 128 × 128 · f64 | 97750 | 97.75 |
| **Matrix multiplication / nalgebra**<br>256 × 256 · 256 × 256 · f64 | 1022417.0000000001 | 1022.4170000000001 |
| **Matrix multiplication / NumPy**<br>256 × 256 · 256 × 256 · f64 | 120834 | 120.834 |

# Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / Kernix**<br>256 × 256 · 256 × 256 · f64 | 845417 | 845.417 |
| **Matrix multiplication / Candle**<br>256 × 256 · 256 × 256 · f64 | 883000 | 883 |
| **Matrix multiplication / faer**<br>256 × 256 · 256 × 256 · f64 | 822750 | 822.75 |
| **Matrix multiplication / ndarray**<br>256 × 256 · 256 × 256 · f64 | 816333 | 816.333 |
| **Matrix multiplication / nalgebra**<br>64 × 64 · 64 × 64 · f32 | 6917 | 6.917 |
| **Matrix multiplication / NumPy**<br>64 × 64 · 64 × 64 · f32 | 1916 | 1.916 |
| **Matrix multiplication / Candle**<br>64 × 64 · 64 × 64 · f32 | 7916 | 7.916 |
| **Matrix multiplication / Kernix**<br>64 × 64 · 64 × 64 · f32 | 10500 | 10.5 |
| **Matrix multiplication / faer**<br>64 × 64 · 64 × 64 · f32 | 8209 | 8.209 |
| **Matrix multiplication / ndarray**<br>64 × 64 · 64 × 64 · f32 | 6959 | 6.959 |

# Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / Kernix**<br>128 × 128 · 128 × 128 · f32 | 52459 | 52.459 |
| **Matrix multiplication / NumPy**<br>128 × 128 · 128 × 128 · f32 | 6125 | 6.125 |
| **Matrix multiplication / nalgebra**<br>128 × 128 · 128 × 128 · f32 | 51208 | 51.208 |
| **Matrix multiplication / Candle**<br>128 × 128 · 128 × 128 · f32 | 52916 | 52.916 |
| **Matrix multiplication / ndarray**<br>128 × 128 · 128 × 128 · f32 | 47834 | 47.834 |
| **Matrix multiplication / faer**<br>128 × 128 · 128 × 128 · f32 | 55500 | 55.5 |
| **Matrix multiplication / NumPy**<br>256 × 256 · 256 × 256 · f32 | 39916 | 39.916 |
| **Matrix multiplication / faer**<br>256 × 256 · 256 × 256 · f32 | 458125 | 458.125 |
| **Matrix multiplication / Kernix**<br>256 × 256 · 256 × 256 · f32 | 372500 | 372.5 |
| **Matrix multiplication / nalgebra**<br>256 × 256 · 256 × 256 · f32 | 384375 | 384.375 |

# Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Matrix multiplication / Candle**<br>256 × 256 · 256 × 256 · f32 | 408125 | 408.125 |
| **Matrix multiplication / ndarray**<br>256 × 256 · 256 × 256 · f32 | 390542 | 390.542 |
| **Batched multiplication / Candle**<br>4 × 16 × 32 · 1 × 32 × 16 · f64 | 3583 | 3.583 |
| **Batched multiplication / NumPy**<br>4 × 16 × 32 · 1 × 32 × 16 · f64 | 1791 | 1.791 |
| **Batched multiplication / Kernix**<br>4 × 16 × 32 · 1 × 32 × 16 · f64 | 2875 | 2.875 |
| **Column-pivoted QR / faer**<br>96 × 32 · f64 | 40167 | 40.167 |
| **Column-pivoted QR / Kernix**<br>96 × 32 · f64 | 73250 | 73.25 |
| **Column-pivoted QR / nalgebra**<br>96 × 32 · f64 | 96375 | 96.375 |
| **Column-pivoted QR / nalgebra**<br>256 × 64 · f64 | 1009125 | 1009.125 |
| **Column-pivoted QR / faer**<br>256 × 64 · f64 | 309791 | 309.791 |

# Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Column-pivoted QR / Kernix**<br>256 × 64 · f64 | 605417 | 605.417 |
| **Minimum-norm least squares / Kernix**<br>64 × 32, rank 16, 4 RHS · f64 | 51250 | 51.25 |
| **Minimum-norm least squares / NumPy**<br>64 × 32, rank 16, 4 RHS · f64 | 81000 | 81 |
| **Minimum-norm least squares / nalgebra**<br>64 × 32, rank 16, 4 RHS · f64 | 69834 | 69.834 |
| **Minimum-norm least squares / Kernix**<br>32 × 64, rank 32, 4 RHS · f64 | 95917 | 95.917 |
| **Minimum-norm least squares / NumPy**<br>32 × 64, rank 32, 4 RHS · f64 | 136125 | 136.125 |
| **Minimum-norm least squares / nalgebra**<br>32 × 64, rank 32, 4 RHS · f64 | 152958 | 152.958 |
| **Thin QR / faer**<br>96 × 32 · f64 | 50541 | 50.541 |
| **Thin QR / NumPy**<br>96 × 32 · f64 | 47500 | 47.5 |
| **Thin QR / nalgebra**<br>96 × 32 · f64 | 43542 | 43.542 |

# Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Thin QR / Kernix**<br>96 × 32 · f64 | 56792 | 56.792 |
| **Thin QR / faer**<br>256 × 64 · f64 | 303167 | 303.167 |
| **Thin QR / Kernix**<br>256 × 64 · f64 | 497916.99999999994 | 497.91699999999994 |
| **Thin QR / nalgebra**<br>256 × 64 · f64 | 356042 | 356.042 |
| **Thin QR / NumPy**<br>256 × 64 · f64 | 364458 | 364.458 |
| **Cholesky factorization / Kernix**<br>64 × 64 · f64 | 22292 | 22.292 |
| **Cholesky factorization / nalgebra**<br>64 × 64 · f64 | 21042 | 21.042 |
| **Cholesky factorization / faer**<br>64 × 64 · f64 | 10334 | 10.334 |
| **Cholesky factorization / NumPy**<br>64 × 64 · f64 | 15083 | 15.083 |
| **Cholesky factorization / faer**<br>128 × 128 · f64 | 49500 | 49.5 |

# Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Cholesky factorization / Kernix**<br>128 × 128 · f64 | 167667 | 167.667 |
| **Cholesky factorization / NumPy**<br>128 × 128 · f64 | 108167 | 108.167 |
| **Cholesky factorization / nalgebra**<br>128 × 128 · f64 | 104000 | 104 |
| **Cholesky factorization / nalgebra**<br>256 × 256 · f64 | 730125 | 730.125 |
| **Cholesky factorization / faer**<br>256 × 256 · f64 | 250708 | 250.708 |
| **Cholesky factorization / Kernix**<br>256 × 256 · f64 | 1322125 | 1322.125 |
| **Cholesky factorization / NumPy**<br>256 × 256 · f64 | 330125 | 330.125 |
| **Record + backward / Kernix**<br>sum(A · W), 32 × 16 · 16 × 8 · f64 | 8042 | 8.042 |
| **Record + backward / Candle**<br>sum(A · W), 32 × 16 · 16 × 8 · f64 | 19416 | 19.416 |
| **Model inference / Kernix**<br>batched normalized/sliced MLP · f64 | 2250 | 2.25 |

# Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Model inference / Candle**<br>batched normalized/sliced MLP · f64 | 5458 | 5.458 |
| **Model recording + forward / Kernix**<br>batched normalized/sliced MLP · f64 | 2875 | 2.875 |
| **Model recording + forward / Candle**<br>batched normalized/sliced MLP · f64 | 5625 | 5.625 |
| **Model backward / Candle**<br>prepared graph → fresh gradients · f64 | 20959 | 20.959 |
| **Model backward / Kernix**<br>prepared graph → fresh gradients · f64 | 3750 | 3.75 |
| **Parameter update / Kernix**<br>transactional SGD, fixed gradients · f64 | 750 | 0.75 |
| **Row softmax / ndarray**<br>256 × 64 · f64 | 65250 | 65.25 |
| **Row softmax / NumPy**<br>256 × 64 · f64 | 76667 | 76.667 |
| **Row softmax / Candle**<br>256 × 64 · f64 | 64999.99999999999 | 64.99999999999999 |
| **Row softmax / Kernix**<br>256 × 64 · f64 | 78917 | 78.917 |

# Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

| WORKLOAD / LIBRARY | NANOSECONDS | MICROSECONDS |
|---|---|---|
| **Create zeros / Kernix**<br>64 × 64 · f32 | 750 | 0.75 |
| **Create zeros / faer**<br>64 × 64 · f32 | 291 | 0.291 |
| **Create zeros / ndarray**<br>64 × 64 · f32 | 291 | 0.291 |
| **Create zeros / NumPy**<br>64 × 64 · f32 | 542 | 0.542 |
| **Create zeros / Candle**<br>64 × 64 · f32 | 458 | 0.458 |
| **Create ones / faer**<br>64 × 64 · f32 | 333 | 0.333 |
| **Create ones / Kernix**<br>64 × 64 · f32 | 292 | 0.292 |
| **Create ones / Candle**<br>64 × 64 · f32 | 417 | 0.417 |
| **Create ones / NumPy**<br>64 × 64 · f32 | 1125 | 1.125 |
| **Create ones / ndarray**<br>64 × 64 · f32 | 333 | 0.333 |